

treatment (recurrence of depression, maternal health, infant bonding) to the mother and child

- For many patients, this may mean continuing treatment during pregnancy
- Exposure to serotonin reuptake inhibitors early in pregnancy may be associated with increased risk of septal heart defects (absolute risk is small)
- Use of serotonin reuptake inhibitors beyond the 20th week of pregnancy may be associated with increased risk of pulmonary hypertension in newborns, although this is not proven
- Exposure to serotonin reuptake inhibitors late in pregnancy may be associated with increased risk of gestational hypertension and preeclampsia
- Neonates exposed to SSRIs or SNRIs late in the third trimester have developed complications requiring prolonged hospitalization, respiratory support, and tube feeding; reported symptoms are consistent with either a direct toxic effect of SSRIs and SNRIs or, possibly, a drug discontinuation syndrome, and include respiratory distress, cyanosis, apnea, seizures, temperature instability, feeding difficulty, vomiting, hypoglycemia, hypotonia, hypertonia, hyperreflexia, tremor, jitteriness, irritability, and constant crying
- National Pregnancy Registry for Antidepressants: 1-866-961-2388, <https://womensmentalhealth.org/research/pregnancyregistry/antidepressants>

Breast Feeding

- Unknown if vortioxetine is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- If child becomes irritable or sedated, breast feeding or drug may need to be discontinued
- Immediate postpartum period is a high-risk time for depression, especially in women who have had prior depressive episodes, so drug may need to be reinstituted late in the third trimester or shortly after childbirth to prevent a recurrence during the postpartum period
- Must weigh benefits of breast feeding with risks and benefits of antidepressant treatment versus nontreatment to both the infant and the mother
- For many patients, this may mean continuing treatment during breast feeding

The Art Of Psychopharmacology

Potential Advantages

- Patients with sexual dysfunction
- Patients with cognitive symptoms of depression
- Patients with residual cognitive symptoms after treatment with another antidepressant
- Elderly patients

- Patients who have not responded to other antidepressants
- Patients who do not want weight gain

Potential Disadvantages

- Cost

Primary Target Symptoms

- Depressed mood
- Cognitive symptoms
- Anxiety

Pearls

- In May 2016 the US FDA approved a brand name change from Brintellix to Trintellix in order to decrease prescribing and dispensing errors due to name confusion with the anti-platelet medication Brilinta (ticagrelor)
- May have less sexual dysfunction than SSRIs
- Multiple studies show pro-cognitive effects greater than a comparator antidepressant in patients with major depressive episodes
- Patients who do not respond to antidepressants with other mechanisms of action may respond to vortioxetine

- Shown effective specifically in elderly patients with depression, with a positive trial in geriatric depression with improvement of cognition as well as mood
- Has a unique claim of preventing recurrences in major depression
- No weight gain in clinical trials
- Long half-life means vortioxetine can generally be abruptly discontinued, although some caution may be necessary when stopping higher doses (i.e., 15 or 20 mg/day)
- Despite serotonin 3 antagonist actions, nausea is common, presumably due to full agonist actions at serotonin 1A receptors
- Dose response for efficacy in depression: higher doses are more effective
- Vortioxetine has a unique multimodal mechanism of action
- Nonresponse to vortioxetine in elderly may require consideration of mild cognitive impairment or Alzheimer disease

Suggested Reading

Bang-Anderson B , Ruhland T , Jorgensen M , et al. Discovery of 1-[2-(2,4-dimethylphenylsulfanyl)phenyl]piperazine (Lu AA21004): a novel multimodal compound for the treatment of major depressive disorder . *J Med Chem* 2011 ;**54** (9):3206 –21.

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Mork A , Montezinho LP , Miller S , et al. Vortioxetine (LU AA21004), a novel multimodal antidepressant, enhanced memory in rats . *Pharmacol Biochem Behav* 2013 ;**105** :41 –50 .
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Westrich I , Pehrson A , Zhong H , et al. In vitro and in vivo effects of the multimodal antidepressant vortioxetine (Lu AA21004) at human and rat targets . *Int J Psychiatry Clin Pract* 2012 ;**5** (Suppl 1):S47 .
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Zaleplon

Therapeutics

Brands

- Sonata
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: GABA positive allosteric modulator (GABA-PAM)
- Non-benzodiazepine hypnotic; alpha 1 isoform agonist of GABA-A/benzodiazepine receptors

Commonly Prescribed for

(bold for FDA approved)

- **Short-term treatment of insomnia**



How the Drug Works

- Binds selectively to a subtype of the benzodiazepine receptor, the alpha 1 isoform

- May enhance GABA inhibitory actions that provide sedative hypnotic effects more selectively than other actions of GABA
- Boosts chloride conductance through GABA-regulated channels
- Inhibitory actions in sleep centers may provide sedative hypnotic effects

How Long Until It Works

- Generally takes effect in less than an hour

If It Works

- Improves quality of sleep
- Effects on total wake-time and number of nighttime awakenings may be decreased over time

If It Doesn't Work

- If insomnia does not improve after 7–10 days, it may be a manifestation of a primary psychiatric or physical illness such as obstructive sleep apnea or restless leg syndrome, which requires independent evaluation
- Increase the dose
- Improve sleep hygiene
- Switch to another agent

Best Augmenting Combos for Partial Response or Treatment Resistance

- Generally, best to switch to another agent
- Trazodone
- Agents with antihistamine actions (e.g., diphenhydramine, TCAs)

Tests

- None for healthy individuals

Side Effects

How Drug Causes Side Effects

- Actions at benzodiazepine receptors that carry over to the next day can cause daytime sedation, amnesia, and ataxia
- Long-term adaptations of zaleplon not well studied, but chronic studies of other alpha 1 selective non-benzodiazepine hypnotics suggest lack of notable tolerance or dependence developing over time

Notable Side Effects

- * Sedation
- * Dizziness, ataxia
- * Dose-dependent amnesia

* Hyperexcitability, nervousness

- Rare hallucinations
- Headache
- Decreased appetite



Life-Threatening or Dangerous Side Effects

- Respiratory depression, especially when taken with other CNS depressants in overdose
- Rare angioedema

Weight Gain



- Reported but not expected

Sedation



- Many experience and/or can be significant in amount

What to Do About Side Effects

- Wait
- To avoid problems with memory, do not take zaleplon if planning to sleep for less than 4 hours
- Lower the dose

- Administer flumazenil if side effects are severe or life-threatening

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing And Use

Usual Dosage Range

- 10 mg/day at bedtime for 7–10 days

Dosage Forms

- Capsule 5 mg, 10 mg

How to Dose

- Initial 10 mg/day at bedtime; may increase to 20 mg/day at bedtime if ineffective; maximum dose generally 20 mg/day



Dosing Tips

- Patients with lower body weights may require only a 5 mg dose
- Zaleplon should generally not be prescribed in quantities greater than a 1-month supply
- Risk of dependence may increase with dose and duration of treatment

* However, treatment with alpha 1 selective non-benzodiazepine hypnotics may cause less tolerance or dependence than benzodiazepine hypnotics

Overdose

- No fatalities reported with zaleplon; fatalities have occurred with other sedative hypnotics; sedation, confusion, ataxia, hypotension, respiratory depression, coma

Long-Term Use

- Not generally intended for long-term use
- Increased wakefulness during the latter part of the night (wearing off) or an increase in daytime anxiety (rebound) may occur because of short half-life

Habit Forming

- Zaleplon is a Schedule IV drug
- Some patients may develop dependence and/or tolerance; risk may be greater with higher doses
- History of drug addiction may increase risk of dependence

How to Stop

- Rebound insomnia may occur the first night after stopping

- If taken for more than a few weeks, taper to reduce chances of withdrawal effects

Pharmacokinetics

- Terminal phase elimination half-life approximately 1 hour (ultra-short half-life)



Drug Interactions

- Increased depressive effects when taken with other CNS depressants
- Cimetidine may increase plasma concentrations of zaleplon, requiring a lower initial dose of zaleplon (5 mg/day)
- CYP450 3A4 inducers such as carbamazepine may reduce the effectiveness of zaleplon



Other Warnings/Precautions

- Insomnia may be a symptom of a primary disorder, rather than a primary disorder itself
- Some patients may exhibit abnormal thinking or behavioral changes similar to those caused by other CNS depressants (i.e., either depressant actions or disinhibiting actions)
- Some depressed patients may experience a worsening of suicidal ideation
- Use only with extreme caution in patients with impaired respiratory function or obstructive sleep apnea

- Zaleplon should be administered only at bedtime
- Rare angioedema has occurred with sedative hypnotic use and could potentially cause fatal airway obstruction if it involves the throat, glottis, or larynx; thus if angioedema occurs treatment should be discontinued
- Sleep driving and other complex behaviors, such as eating and preparing food and making phone calls, have been reported in patients taking sedative hypnotics; in some cases these behaviors have resulted in serious injury or death, prompting the FDA to require a black box warning

Do Not Use

- If the patient has experienced an episode of complex sleep behaviors after taking a sleep medication
- If there is a proven allergy to zaleplon

Special Populations

Renal Impairment

- No dose adjustment necessary
- Use with caution in patients with severe impairment

Hepatic Impairment

- Mild to moderate impairment: recommended dose 5 mg

- Not recommended for use in patients with severe impairment

Cardiac Impairment

- Zaleplon has not been studied in patients with cardiac impairment, but dose adjustment may not be necessary

Elderly

- Recommended dose: 5 mg



Children and Adolescents

- Safety and efficacy have not been established
- Long-term effects of zaleplon in children/adolescents are unknown
- Should generally receive lower doses and be more closely monitored



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women

- In animal studies, administration during organogenesis was not teratogenic at doses up to 49 (rats) and 48 (rabbits) times the maximum recommended human dose (MRHD); in rats, pre- and postnatal growth was reduced and maternal toxicity occurred
- In a pre- and postnatal development study in rats, increased stillbirth and postnatal mortality, and decreased growth and physical development, were observed when zaleplon was administered at doses of 7 mg/kg/day during the latter part of gestation and during lactation; the no-effect dose for offspring development was 1 mg/kg/day (0.5 times the MRHD of 20 mg on a mg/m² basis)
- Infants whose mothers took sedative hypnotics during pregnancy may experience some withdrawal symptoms
- Neonatal flaccidity has been reported in infants whose mothers took sedative hypnotics during pregnancy

Breast Feeding

- Some drug is found in mother's breast milk
- * Recommended either to discontinue drug or bottle feed

The Art Of Psychopharmacology

Potential Advantages

- Those needing short duration of action

Potential Disadvantages

- Those needing longer duration of action
- More expensive than some other sedative hypnotics

Primary Target Symptoms

- Time to sleep onset
- Total sleep time
- Nighttime awakenings

Pearls

- Zaleplon has not been shown to increase the total time asleep or to decrease the number of awakenings
- * May be preferred over benzodiazepines because of its rapid onset of action, short duration of effect, and safety profile
- * Popular for uses requiring short half-life (e.g., dosing in the middle of the night, sleeping on airplanes, jet lag)
- * May not be ideal for patients who desire immediate hypnotic onset and eat just prior to bedtime
- Not a benzodiazepine itself, but binds to benzodiazepine receptors
- May have fewer carryover side effects than some other sedative hypnotics
- May not have sufficient efficacy in patients with severe chronic insomnia resistant to some other sedative hypnotics

- May cause less dependence than some other sedative hypnotics, especially in those without a history of substance abuse
- * Zaleplon is not absorbed as quickly if taken with high-fat foods, which may reduce onset of action

Suggested Reading

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Mangano RM . Efficacy and safety of zaleplon at peak plasma levels . *Int J Clin Pract Suppl* 2001 ;**116** :9 –13.

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Weitzel KW , Wickman JM , Augustin SG , Strom JG . Zaleplon: a pyrazolopyrimidine sedative-hypnotic agent for the treatment of insomnia . *Clin Ther* 2000 ;**22** :1254 –67.

Ziprasidone

Therapeutics

Brands

- Geodon
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: dopamine and serotonin receptor antagonist (DS-RAn)
- Atypical antipsychotic (serotonin-dopamine antagonist; second-generation antipsychotic; also a mood stabilizer)

Commonly Prescribed for

(bold for FDA approved)

- **Schizophrenia**
- **Delaying relapse in schizophrenia**
- **Acute agitation in schizophrenia (intramuscular)**
- **Acute mania/mixed mania**

- **Bipolar maintenance**
- Other psychotic disorders
- Bipolar depression
- Behavioral disturbances in dementias
- Behavioral disturbances in children and adolescents
- Disorders associated with problems with impulse control
- Posttraumatic stress disorder

How the Drug Works

- Blocks dopamine 2 receptors, reducing positive symptoms of psychosis and stabilizing affective symptoms
 - Blocks serotonin 2A receptors, causing enhancement of dopamine release in certain brain regions and thus reducing motor side effects and possibly improving cognitive and affective symptoms
 - Interactions at a myriad of other neurotransmitter receptors may contribute to ziprasidone's efficacy
- * Specifically, interactions at 5HT2C and 5HT1A receptors may contribute to efficacy for cognitive and affective symptoms in some patients
- * Specifically, interactions at 5HT1D and 5HT7 receptors and at serotonin and norepinephrine transporters (especially at high doses) may contribute to efficacy for affective symptoms in some patients

How Long Until It Works

- Psychotic and manic symptoms can improve within 1 week, but it may take several weeks for full effect on behavior as well as on cognition and affective stabilization
- Classically recommended to wait at least 4–6 weeks to determine efficacy of drug, but in practice some patients require up to 16–20 weeks to show a good response, especially on negative or cognitive symptoms
- IM formulation can reduce agitation in 15 minutes

If It Works

- Most often reduces positive symptoms in schizophrenia but does not eliminate them
- Can improve negative symptoms, as well as aggressive, cognitive, and affective symptoms in schizophrenia
- Most schizophrenic patients do not have a total remission of symptoms but rather a reduction of symptoms by about a third
- Perhaps 5–15% of schizophrenic patients can experience an overall improvement of greater than 50–60%, especially when receiving stable treatment for more than a year
- Such patients are considered super-responders or “awakeners” since they may be well enough to be employed, live independently, and sustain long-term relationships

- Many bipolar patients may experience a reduction of symptoms by half or more
- Continue treatment until reaching a plateau of improvement
- After reaching a satisfactory plateau, continue treatment for at least a year after first episode of psychosis
- For second and subsequent episodes of psychosis, treatment may need to be indefinite
- Even for first episodes of psychosis, it may be preferable to continue treatment indefinitely to avoid subsequent episodes
- Treatment may not only reduce mania but also prevent recurrences of mania in bipolar disorder

If It Doesn't Work

- Try one of the other atypical antipsychotics
- If 2 or more antipsychotic monotherapies do not work, consider clozapine
- Some patients may require treatment with a conventional antipsychotic
- If no first-line atypical antipsychotic is effective, consider higher doses or augmentation with valproate or lamotrigine
- Consider noncompliance and switch to another antipsychotic with fewer side effects or to an antipsychotic that can be given by depot injection

- Consider initiating rehabilitation and psychotherapy such as cognitive remediation
- Consider presence of concomitant drug abuse

Best Augmenting Combos for Partial Response or Treatment Resistance

- Valproic acid (valproate, divalproex, divalproex ER)
- Other mood-stabilizing anticonvulsants (carbamazepine, oxcarbazepine, lamotrigine)
- Topiramate
- Lithium
- Benzodiazepines

Tests

Before starting an atypical antipsychotic

- * Weigh all patients and track BMI during treatment
- Get baseline personal and family history of diabetes, obesity, dyslipidemia, hypertension, and cardiovascular disease
- * Get waist circumference (at umbilicus), blood pressure, fasting plasma glucose, and fasting lipid profile
- Determine if the patient
 - is overweight (BMI 25.0–29.9)

- is obese (BMI >30)
- has pre-diabetes (fasting plasma glucose 100–125 mg/dL)
- has diabetes (fasting plasma glucose >126 mg/dL)
- has hypertension (BP >140/90 mmHg)
- has dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL cholesterol)
- Treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management

Monitoring after starting an atypical antipsychotic

- * BMI monthly for 3 months, then quarterly
- * Consider monitoring fasting triglycerides monthly for several months in patients at high risk for metabolic complications and when initiating or switching antipsychotics
- * Blood pressure, fasting plasma glucose, fasting lipids within 3 months and then annually, but earlier and more frequently for patients with diabetes or who have gained >5% of initial weight
- Treat or refer for treatment and consider switching to another atypical antipsychotic for patients who become overweight, obese, pre-diabetic, diabetic, hypertensive, or dyslipidemic while receiving an atypical antipsychotic

* Even in patients without known diabetes, be vigilant for the rare but life-threatening onset of diabetic ketoacidosis, which always requires immediate treatment, by monitoring for the rapid onset of polyuria, polydipsia, weight loss, nausea, vomiting, dehydration, rapid respiration, weakness, and clouding of sensorium, even coma

- Routine ECGs for screening or monitoring of dubious clinical value
- ECGs may be useful for selected patients (e.g., those with personal or family history of QTc prolongation; cardiac arrhythmia; recent myocardial infarction; uncompensated heart failure; or those taking agents that prolong QTc interval such as pimozide, thioridazine, selected antiarrhythmics, moxifloxacin, sparfloxacin, etc.)
- Patients at risk for electrolyte disturbances (e.g., patients on diuretic therapy) should have baseline and periodic serum potassium and magnesium measurements
- Patients with low white blood cell count (WBC) or history of drug-induced leukopenia/neutropenia should have complete blood count (CBC) monitored frequently during the first few months and ziprasidone should be discontinued at the first sign of decline of WBC in the absence of other causative factors
- Patients should be monitored periodically for the development of abnormal movements of tardive dyskinesia, using neurological exam and the AIMS (Abnormal Involuntary Movement Scale)

Side Effects

How Drug Causes Side Effects

- Blocking alpha 1 adrenergic receptors can cause dizziness, hypotension, and syncope, especially at high doses
- Acute blockade of dopamine 2 receptors in the striatum can cause drug-induced parkinsonism, dystonia, or akathisia
- Chronic blockade of dopamine 2 receptors in the striatum can cause tardive dyskinesia
- By blocking dopamine 2 receptors in the pituitary, it can cause elevations in prolactin
- Mechanism of weight gain and increased incidence of diabetes and dyslipidemia with atypical antipsychotics is unknown, and risks vary based on the particular agent

Notable Side Effects

- * Activation (at very low to low doses)
- Dizziness, drug-induced parkinsonism, sedation (dose-dependent), dystonia at high doses
- Nausea, dry mouth (dose-dependent)
- Asthenia, skin rash
- Orthostatic hypotension (dose-dependent)
- Tardive dyskinesia (reduced risk compared to conventional antipsychotics)